Employment Equity Data Analysis (2013–2024)

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Abstract

This report presents an exploratory analysis of employment equity data covering the period 2013 to 2024. The analysis uses two datasets: an employer metadata file containing registration and organisational information, and a workforce profile file containing employee counts by occupational level, gender, and demographic group. The main objectives were to assess data quality, clean and integrate the datasets, examine changes in workforce gender composition, and investigate women's representation in top management across industries.

The original metadata dataset contained 500 employer references and 73 columns, while the workforce profile dataset contained 42,000 records and 13 columns. The metadata was transformed from a wide format into a year-specific structure and joined to the workforce profile using employer reference and reporting year. All 42,000 workforce records matched an employer-year metadata record. Duplicate employer-year-occupation keys, missing workforce count values, and negative workforce counts were not identified in the profile data. Inconsistent occupational labels were standardised before analysis.

The results show that women's share of the reported workforce increased from 47.21% in 2013 to 53.79% in 2024. Women's representation in top management also increased, from 17.81% to 24.70% over the same period. However, the 2024 industry analysis revealed substantial differences. Among industries represented by at least ten employer references, women's share of top-management positions ranged from 12.94% in construction to 51.43% in education. The analysis also identified missing employer names and industry-classification concerns that require further investigation.

These findings describe the supplied employer records and should not automatically be interpreted as representative of all South African employers. The report concludes that consistent data validation, improved demographic reporting, and closer monitoring of senior leadership representation would strengthen employment equity analysis.

Chapter 1

Introduction

Employment equity is concerned with promoting fair access to employment opportunities and improving representation across different groups within the workforce. Examining workforce composition over time can help identify changes in representation and highlight areas where inequalities may persist, particularly in senior management positions.

Data analysis provides a practical way to examine these patterns. By organising workforce records according to reporting year, occupational level, gender, demographic group, and industry, it becomes possible to compare representation across different periods and sectors. However, meaningful conclusions depend on the quality, consistency, and structure of the underlying data.

This report analyses two employment equity datasets covering the period from 2013 to 2024. The first dataset contains employer metadata, including registration names, industry sectors, business types, provincial locations, towns, and SETA classifications. The second contains workforce counts for different occupational levels, disaggregated by gender and demographic group.

The analysis focuses on four main objectives:

1. Assess the structure, completeness, consistency, and quality of the supplied datasets.
2. Clean and restructure the data and combine the employer metadata with the workforce profiles.
3. Examine changes in the gender composition of the reported workforce and women's representation in top management between 2013 and 2024.
4. Compare women's representation in top management across industries and identify data-quality issues that may affect interpretation.

The analysis is exploratory and descriptive. It identifies patterns in the supplied records but does not establish the causes of those patterns or prove that the records represent the entire South African workforce.

1.1 Data and Preparation

1.1.1 Data sources and structure

Two CSV files were supplied for the analysis.

The first file, `ee_meta.csv`, contains 500 employer references and 73 columns. The columns include an employer reference identifier and year-specific information about registration names, industry sectors, business types, physical provinces, towns, and SETA classifications. The metadata is stored in a wide format, where information for different reporting years appears in separate columns.

The second file, `ee_profile_all.csv`, contains 42,000 records and 13 columns. Each record represents an employer's workforce profile for a particular reporting year and occupational level. The profile includes the reporting year, employer reference, occupational level, and ten workforce-count columns representing female and male employees across the demographic groups recorded in the source data.

The workforce profile covers seven occupational levels and twelve reporting years, from 2013 to 2024. Each year contains 3,500 records, corresponding to 500 employer references and seven occupational levels per employer.

The following table summarises the supplied datasets.

Table 1.1: Overview of the supplied datasets

Characteristic	**Employer metadata**	**Workforce profile**
Number of rows	500	42,000
Number of columns	73	13
Reporting period	2013–2024	2013–2024
Main purpose	Employer characteristics	Workforce composition
Original structure	Wide	Long

The two files complement one another: the workforce profile provides employee counts, while the metadata provides the employer characteristics needed to compare workforce representation across industries and reporting years.

1.2 Data cleaning and standardisation

The original files were retained, and data preparation was carried out on working copies. The first stage involved inspecting the dimensions, column names, data types, reporting years, missing values, and occupational-level labels.

The metadata required restructuring because year-specific attributes were stored in separate columns. For example, employer industry information was recorded separately for 2013 and 2024. These columns were transformed from a wide format into a long format, with the reporting year represented as a separate variable. The resulting dataset was organised so that each employer reference had one metadata record for each reporting year.

The workforce profile contained inconsistent capitalisation and spelling in some demographic column names. These names were mapped explicitly to their corresponding demographic groups and genders to ensure that female and male counts were aggregated consistently. The original column names were not treated as interchangeable without checking their meaning.

An inconsistency was also found in the occupational-level labels. The label `Temporary` appeared in some records, whereas `Temporary employees` appeared in others. These labels were standardised to `Temporary employees` so that the same occupational category would not be counted as two separate groups.

The cleaning process also included checking for duplicate records, missing workforce counts, and negative values. No duplicate employer-year-occupation keys, missing workforce count values, or negative workforce counts were identified in the supplied workforce profile.

1.3 Integration of the datasets

The reshaped metadata and workforce profile were combined using the employer reference (`ee_ref`) and reporting year (`submission_year`) as joint keys. Using both keys was important because an employer's characteristics may differ across reporting years.

A left join was performed from the workforce profile to the employer metadata. All 42,000 workforce records matched a corresponding employer-year metadata record. This confirmed that every workforce profile record had a matching metadata entry for the same employer reference and reporting year.

The resulting integrated dataset retained the workforce counts alongside employer characteristics, including industry sector and physical location. This structure supported analysis by year, occupational level, gender, demographic group, and industry.

1.4 Analytical approach

The analysis used descriptive statistics and percentage calculations to examine workforce representation. Female and male workforce totals were calculated by summing the corresponding counts across the demographic groups recorded in the dataset.

The female share of the reported workforce was calculated as

$$\text{Female share (\%)} = \frac{\text{Total female workforce}}{\text{Total reported workforce}} \times 100,$$

where the total reported workforce is the sum of the female and male counts.

Women's representation in top management was calculated separately using records classified as *Top management*. This distinction was necessary because overall workforce representation does not necessarily reflect representation in senior leadership.

For the industry comparison, the 2024 top-management records were grouped by industry sector. To avoid drawing conclusions from industries represented by very few employers, the comparison included only industries with at least ten distinct employer references. Missing or potentially inconsistent classifications were treated as data-quality concerns rather than automatically corrected.

The results were presented using summary tables and graphical visualisations to make changes over time and differences between industries easier to interpret.

Chapter 2

Findings and Discussion

2.1 Gender composition of the reported workforce

The analysis indicates a substantial change in the gender composition of the reported workforce between 2013 and 2024. Women's share increased from 47.21% in 2013 to 53.79% in 2024, while men's share decreased from 52.79% to 46.21%.

Women accounted for less than half of the reported workforce in 2013. Their share approached parity in 2019, when women represented approximately 49.91% and men 50.09%. Women subsequently accounted for more than half of the reported workforce from 2020 onwards.

The female share reached 56.48% in 2022 before decreasing to 55.20% in 2023 and 53.79% in 2024. Therefore, although the overall change between the first and final years was positive, the trend was not a continuous increase. The decline after 2022 highlights the importance of examining annual fluctuations rather than relying only on the first and final observations. Further analysis of occupational levels is necessary to determine whether the overall trend is also reflected in senior and top-management positions.

Table 2: Gender Composition of the Reported Workforce, 2013–2024

Year	Female share (%)	Male share (%)
2013	47.206680	52.793320
2014	47.319083	52.680917
2015	48.481346	51.518654
2016	48.937285	51.062715
2017	48.919813	51.080187
2018	49.228810	50.771190
2019	49.912486	50.087514
2020	51.699885	48.300115
2021	52.998609	47.001391
2022	56.481116	43.518884
2023	55.197739	44.802261
2024	53.789095	46.210905

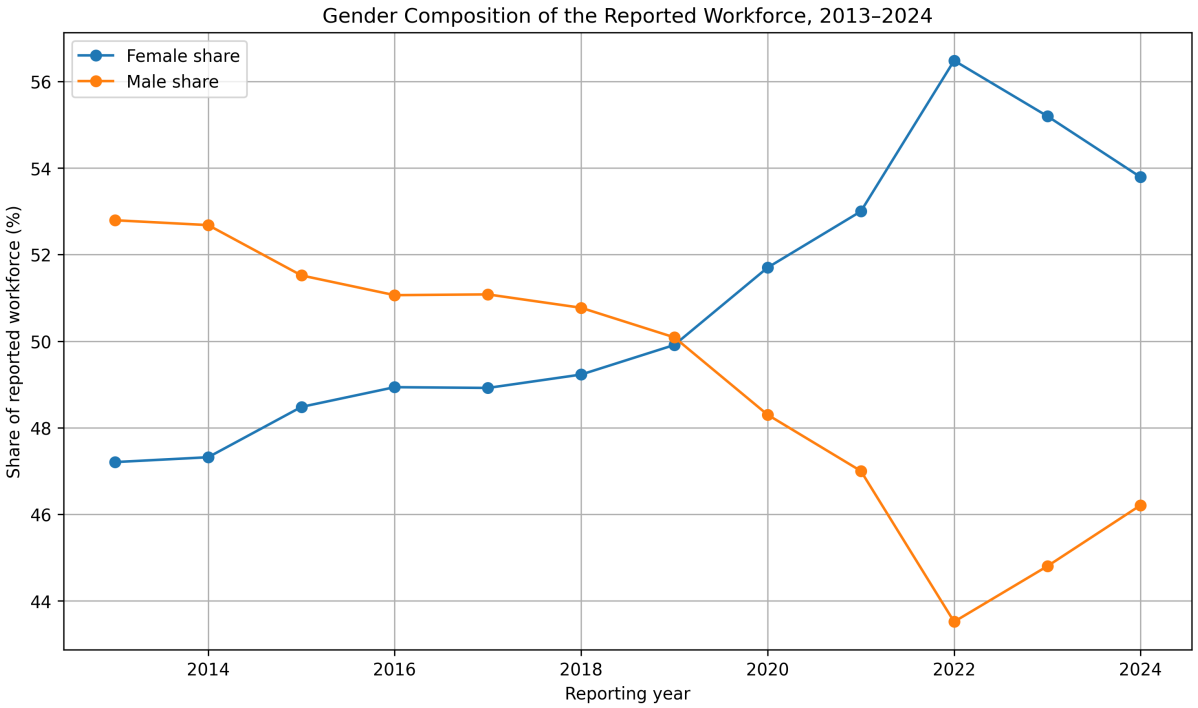


Figure 2.1: Gender composition of the reported workforce, 2013–2024.

Figure 2.1 illustrates the annual changes in female and male workforce shares. The two lines move in opposite directions because the reported female and male shares sum to 100%. The figure provides a clear overview of the shift in the gender composition of the supplied records.

These results describe the employees represented in the dataset. They should not be interpreted as proof that the gender composition of the entire South African workforce changed by the same amount.

2.2 Women's representation in top management

Although the overall workforce became more female-representative, women's representation in top management remained substantially lower than their share of the workforce.

The analysis of women's representation in top-management positions between 2013 and 2024 indicates a gradual improvement in female representation. In 2013, women occupied 17.81% of top-management positions, representing 295 out of 1,656 positions. This proportion increased over the study period, reaching 20.98% in 2018 and 22.71% in 2020. Although a slight decline occurred in 2021, when women's representation decreased from 22.71% to 22.70%, the overall trend remained upward. By 2024, women occupied 24.70% of top-management positions, representing 374 out of 1,514 positions.

Between 2013 and 2024, women's representation in top management increased by approximately 6.89 percentage points. However, women still held fewer than one-quarter of top-management positions in 2024. This finding suggests that the improvement in women's representation across the reported workforce has not yet translated into equal representation at the highest occupational level. The results highlight the importance of examining gender representation across different occupational levels rather than relying solely on overall workforce percentages.

Table: Gender Representation in Top Management, 2013–2024

Year	**Women**	**Men**	**Total**	**Women** (%)
2013	295	1361	1656	17.814010
2014	296	1359	1655	17.885196
2015	315	1318	1633	19.289651
2016	340	1329	1669	20.371480
2017	341	1318	1659	20.554551
2018	344	1296	1640	20.975610
2019	349	1279	1628	21.437346
2020	359	1222	1581	22.707147
2021	357	1216	1573	22.695486
2022	362	1189	1551	23.339781
2023	373	1176	1549	24.080052
2024	374	1140	1514	24.702774

This difference is important because overall workforce representation and representation in leadership measure different aspects of employment equity. A workforce may have a relatively balanced gender composition while senior positions remain disproportionately occupied by men.

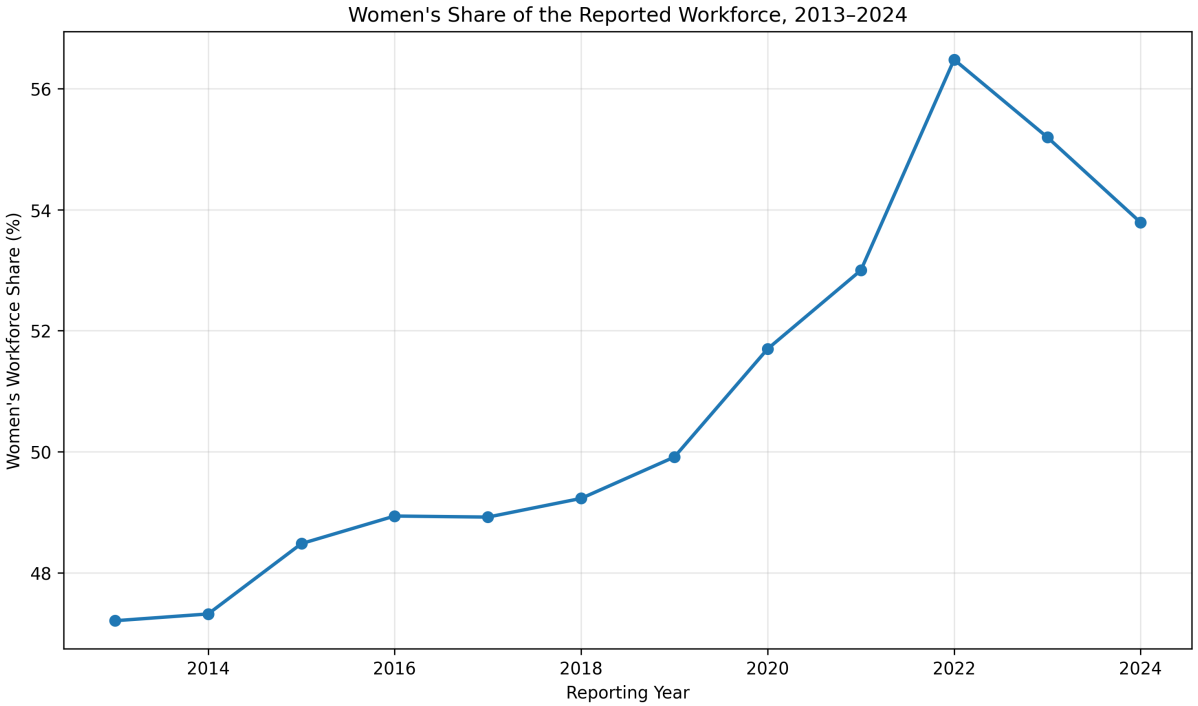


Figure 2.2: Women's share of reported top-management positions, 2013–2024.

Figure 2.2 shows the increase in women's share of top management over the reporting period. The upward trend is encouraging, but the 2024 result indicates that a substantial gender representation gap remained in the supplied records.

2.2.1 Women's representation in top management by industry in 2024

The industry comparison reveals considerable variation in women's representation in top management. The analysis included industries with at least ten distinct employer references in 2024, reducing the influence of results based on very small numbers of employers.

Construction recorded the lowest female share of top management among the included industries, at 12.94%. Agriculture, Forestry and Fishing followed at 13.13%. These results indicate that women occupied a relatively small proportion of reported top-management positions in these two industries.

At the other end of the comparison, Education recorded a female top-management share of 51.43%. Professional, Scientific and Technical Activities recorded 41.51%, while Human Health and Social Work Activities recorded 38.46%.

The difference between construction and education was 38.49 percentage points. This contrast demonstrates that women's representation in senior leadership varied considerably across the industries represented in the dataset.

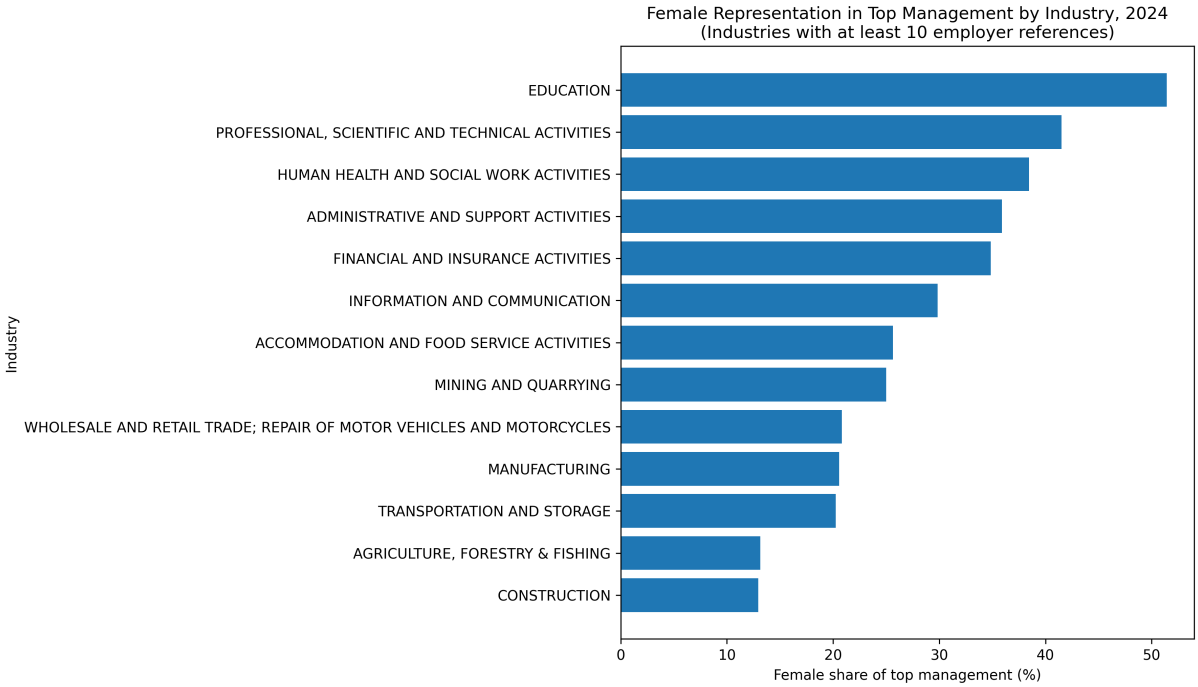


Figure 2.3: Female representation in top management by industry in 2024. Only industries with at least ten employer references are included.

Figure 2.3 compares women's share of top-management positions across the selected industries. The differences may help identify areas for further investigation and monitoring. However, the descriptive results do not establish why the differences exist. Additional information about recruitment, promotion, qualifications, workforce composition, and organisational practices would be required to investigate possible explanations.

2.2.2 Workforce demographic composition

The demographic breakdown also indicates changes in the composition of the reported workforce. The African demographic group accounted for approximately 74.12% of the reported workforce in 2013 and 79.52% in 2024. Over the same period, the White demographic group's share decreased from approximately 10.68% to 6.86%.

These percentages were calculated by combining the female and male counts within each demographic group and dividing the resulting group total by the total reported workforce for the corresponding year.

The results describe the composition of the supplied records and do not, by themselves, explain the reasons for changes in demographic shares. Interpretation should also account for possible changes in the employers represented, the completeness of reporting, and the consistency of demographic classifications across years.

2.2.3 Female Representation in the Overall Workforce and Top Management

The analysis examined changes in female representation in the overall workforce and top management between 2013 and 2024. The results indicate an increase in women's representation across both categories, although the level of representation differed considerably between the overall workforce and top management.

Female representation in the overall workforce increased from 47.21% in 2013 to 53.79% in 2024, representing an increase of 6.58 percentage points. The proportion increased gradually between 2013 and 2019, reaching 49.91%, before exceeding 50% in 2020. Female representation subsequently peaked at 56.48% in 2022, before declining to 55.20% in 2023 and 53.79% in 2024. Although the overall trend indicates an increase over the period, the decline after 2022 suggests that the highest level of female representation was not sustained.

A similar upward trend was observed in top management. Female representation increased from 17.81% in 2013 to 24.70% in 2024, an improvement of 6.89 percentage points. Despite this progress, women accounted for less than one-quarter of top-management employees in 2024. This indicates that improvements in women's representation at the highest organisational level remained limited relative to their representation across the overall workforce.

Table 2.1: Female representation in the overall workforce and top management, selected years

Year	Overall workforce (%)	Top management (%)	Gap (pp)
2013	47.21	17.81	29.39
2016	48.94	20.37	28.57
2019	49.91	21.44	28.48
2020	51.70	22.71	28.99
2022	56.48	23.34	33.14
2024	53.79	24.70	29.09

Note: The representation gap is the difference between female representation in the overall workforce and top management, measured in percentage points (pp).

The difference between female representation in the overall workforce and top management further illustrates this disparity. The representation gap declined slightly from 29.39 percentage points in 2013 to 29.09 percentage points in 2024. However, the gap was not consistently narrowing throughout the period. It reached a maximum of 33.14 percentage points in 2022, when female representation in the overall workforce peaked at 56.48%, compared with 23.34% in top management. By 2024, the gap had decreased, but remained substantial.

Overall, the findings suggest that increased female representation in the overall workforce was not accompanied by equivalent representation in top management. While the data show improvements in both categories, the persistent representation gap highlights the difference between women's participation in the workforce and their presence in the highest organisational positions. These findings are descriptive and do not establish the factors responsible for the observed differences. Furthermore, the results should be interpreted in the context of the supplied dataset and should not automatically be generalised to the entire South African labour market.

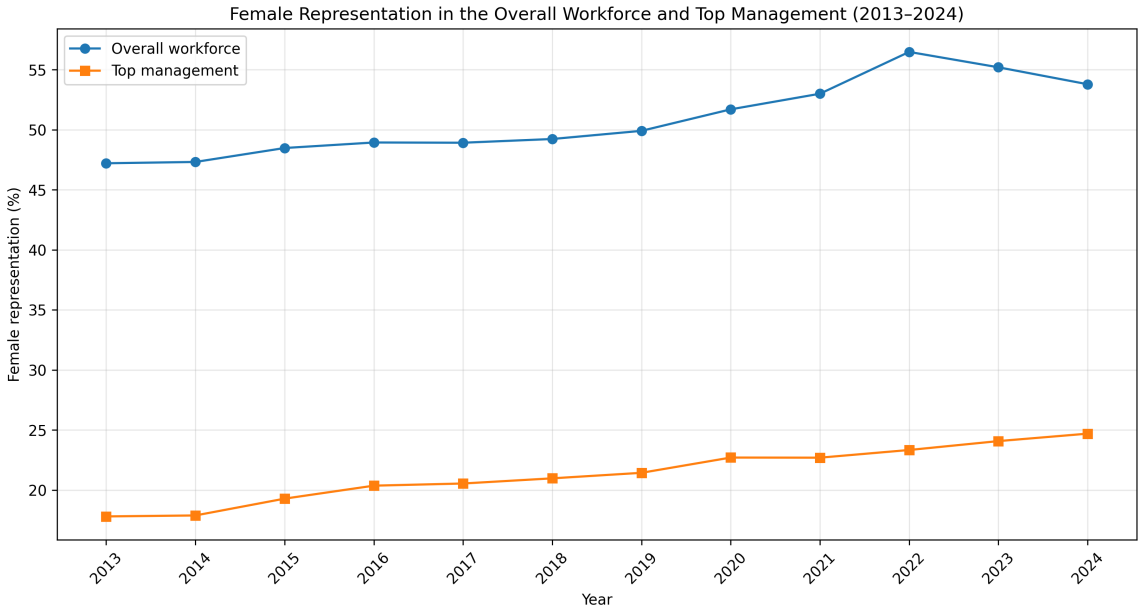


Figure 2.4: Female representation in the overall workforce and top management, 2013–2024.

2.3 Data quality and limitations

Several data-quality considerations affect the interpretation of the results.

First, employer registration names were missing for 29 of the 500 employer references in the 2024 metadata. Six industry-sector values were missing in the 2022 metadata. Missing names may limit the identification and verification of individual employers, while missing industry values may affect comparisons between sectors.

Second, the metadata contained an employer record for the KwaZulu-Natal Provincial Department of Education classified under Arts, Entertainment and Recreation in 2024. Its reported workforce total was 130,309, approximately 36.5% of the total workforce recorded for that year. Given the organisation's name and the size of its reported workforce, the classification should be verified against an authoritative source before drawing industry-

level conclusions. The analysis did not automatically change the classification because the supplied data alone does not establish the correct alternative.

Third, the total reported workforce varied across years. The summed count was 384,077 in 2013, reached 420,190 in 2022, and was 356,985 in 2024. These totals should be interpreted cautiously because the analysis does not establish whether the employer sample and reporting coverage remained fully comparable across the entire period.

Finally, the dataset contains 500 employer references per year, but this does not establish that the employers constitute a representative sample of all South African employers. The findings therefore apply to the supplied records. They should not automatically be generalised to the national workforce.

Despite these limitations, the absence of duplicate employer-year-occupation keys, missing workforce count values, and negative workforce counts provides a useful basis for descriptive analysis. The remaining concerns relate primarily to metadata completeness, classification consistency, and the representativeness of the supplied records.

Chapter 3

Recommendations

Based on the findings, the following recommendations are proposed:

1. **Improve metadata completeness.** Missing employer names and industry classifications should be investigated and corrected using verified source information where available.

2. **Validate industry classifications.** The classification of unusually large employers, particularly the KwaZulu-Natal Provincial Department of Education record, should be checked before finalising industry comparisons.

3. **Monitor representation in top management.** Overall workforce gender shares should be assessed alongside representation in senior leadership. This would provide a more complete view of gender representation across occupational levels.

4. **Investigate industry differences.** Industries with low female representation in top management, including construction and agriculture, forestry and fishing, warrant further investigation. The descriptive results can guide questions for additional analysis but cannot independently identify the causes of these differences.

5. **Improve year-to-year comparability.** Future analyses should examine whether changes in employer coverage, reporting practices, or workforce composition affect comparisons across reporting years.

6. **Maintain a reproducible analysis process.** The cleaning rules, data transformations, calculations, and visualisations should be documented in code. This would make the results easier to reproduce, audit, and update when new data becomes available.

Chapter 4

Conclusion

This report analysed employment equity data covering the period 2013 to 2024 by cleaning, restructuring, and integrating employer metadata with workforce profiles. The preparation process produced a combined dataset in which all 42,000 workforce records matched an employer reference and reporting year. The analysis also standardised inconsistent occupational labels and checked the workforce counts for missing values, negative values, and duplicate employer-year-occupation keys.

The results show that women's share of the reported workforce increased from 47.21% in 2013 to 53.79% in 2024. Women's share of top-management positions also increased, from 17.81% to 24.70%. Nevertheless, the difference between women's overall workforce representation and their representation in top management indicates that the two measures should be examined separately.

The 2024 industry comparison showed that women's representation in top management differed considerably between industries. Among industries with at least ten employer references, the female share ranged from 12.94% in construction to 51.43% in education. The demographic analysis also showed changes in the composition of the reported workforce between 2013 and 2024.

The findings must be interpreted in light of missing metadata, a potentially inconsistent industry classification, variations in reported workforce totals, and uncertainty about the representativeness of the employer records. These limitations do not remove the value of the analysis, but they emphasise the importance of validating source information before making broader claims.

Overall, the analysis demonstrates how data cleaning, integration, descriptive statistics, and visualisation can be used to identify workforce representation patterns and areas for further investigation. More complete metadata, validated classifications, and consistent reporting would improve future employment equity analyses and support more reliable comparisons over time.

Chapter 5

Reproducibility and Supporting Materials

The analysis was conducted in Python using pandas for data manipulation and Matplotlib for visualisation. The supporting code documents the data-cleaning procedures, integration of the datasets, calculation of summary statistics, and generation of figures.

The analysis code and supporting materials are available at: `https://github.com/malose421/71point4-employment-equity-analysis_2/tree/main`